

Recent advances in Mineral and Bone Disorders in CKD

Henk van den Broek & Jonathan Elliott, London, UK

Hyperphosphatemia is a well-known risk factor for disease progression and survival in cats and dogs with chronic kidney disease (CKD), and multiple studies have shown the beneficial effects of reducing the serum phosphate concentration by feeding phosphate-restricted clinical renal diets to these animals (Elliott et al., 2000; Jacob et al., 2002; Plantinga et al., 2005; Ross et al., 2006; Boyd et al., 2008; Chakrabarti et al., 2012). Reduced capacity of diseased kidneys to excrete sufficient phosphate on a daily basis as a consequence of CKD is thought to underlie the biochemical abnormalities of calcium-phosphate homeostasis, bone disease, and soft tissue calcification that are grouped together as chronic kidney disease-mineral and bone disorder (CKD-MBD) (Moe et al., 2006; Slatopolsky 2011). For decades the disturbances in calcium-phosphate homeostasis observed in patients with CKD and its consequences have been explained by hyperphosphatemia and (patho)-physiological increases in parathyroid hormone (PTH), i.e. secondary renal hyperparathyroidism, but recent advances in the understanding of phosphate homeostasis have been made by the discoveries of the phosphaturic hormone fibroblast growth factor 23 (FGF23) and its co-receptor, α -Klotho. Recognition of the role of magnesium in CKD-MBD is another recent breakthrough in the field of nephrology.

Fibroblast growth factor 23	+
-----------------------------	---

FGF23 and CKD IRIS Stage	+
--------------------------	---

FGF23 and hyperphosphatemia	+
-----------------------------	---

FGF23 and survival	+
--------------------	---

Measuring FGF23

+

Magnesium

+

Magnesium and CKD IRIS stage

+

Magnesium and FGF23

+

Magnesium and survival

+

Measuring magnesium

+

Bone disease and soft tissue calcification

+

References

+

BACK

